

MAINS MASTER PROGRAM (MMP) 2024

ECONOMY-5;

AGRICULTURE-3

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2. FARM MECHANIZATION

1) QUESTION-1

- Question1:

- Discuss the significance of agriculture mechanization in enhancing agricultural productivity and rural development in India. Highlight the key challenges associated with the adoption of mechanized farming practices [10 marks, 150 words]

- Introduction

- Farm/Agri mechanization is the process of replacing human and animal labour with machines in agriculture sector. The use of tractors, threshers, harvesters, pump sets etc. are all steps towards farm mechanization.

- Advantages of Farm Mechanization-

- **Increased Productivity** and thus **improved farm income**
 - Various surveys show that productivity per hectare is comparatively higher in a mechanized farm than that of non-mechanized farm
 - It also reduces post harvest losses
 - It cuts down production cost in long run. This would not only contribute to reduced food inflation but can also contribute to increasing competitiveness of India's agri exports.
 - Mechanized deep ploughing increases **soil fertility** and helps in **controlling weed**.
 - **Increasing area under cultivation**: It can also make many fallow not cultivable land cultivable.
- **Contribute to Sustainability**:
 - Better Water Use Efficiency with the help of drip irrigation, sprinklers etc.
 - Efficient stubble management is possible with mechanized solutions like happy seeders, inversion ploughing etc.
- Solves the **problem of labor bottlenecks** during the peak period.
 - Further, as per the World Bank estimates percentage of agricultural workers of the total work force would drop to 25.7 percent by 2050 from 58.7% in 2001. Farm mechanization should be able to solve the future shortage of work force in agriculture sector.
- **Skilled Employment**: It has the potential to create skilled jobs in rural areas. Under employed people can be moved away from manual agriculture labor to skilled work with proper skill training intervention.

- Some Limitations/disadvantages of farm mechanizations

- Farm mechanization may reduce employment opportunities in agri-sector.
- It will increase fossil fuel consumption thus causing pollution, green house emission etc.

- Status of farm mechanization in India

- As per NABARD, penetration of mechanization in farm activities is between 40-45%. Countries like USA (95%) Western Europe (95%), Brazil (75%), and China (57%) have higher level of mechanization.
- But still there are a **number of factors which hinder mechanization**
 - » **Small and Fragmented landholdings**

- » **High Cost of Machinery**
- » **Lack of access to Credit** especially long term credit, for small scale farmers hinder their ability to acquisition this machinery.
- » **Infrastructural Challenges** like poor roads, lack of power supply, and limited storage and processing facilities hamper the effective utilization of farm equipment.
- » **Poor Technological Awareness and skill gaps** about operating and repairing machinery acts as a barrier to adoption.
- » **Availability of abundant unskilled and affordable labor force** also discourage farmers from investing in mechanization, especially when the cost benefit ratio is not favorable.
- » **Regional and Crop Specific Variations:** The diversity of crop types and agro climatic conditions necessitates the availability of region specific and crop-specific mechanization solutions, which may not always be readily available.
- » **Limited Research and Development (R&D):** Insufficient research and development activities, related to agriculture mechanization hinder the development of suitable machinery for diverse farming needs.

- **Conclusion1:**

- To increase the penetration of agri-mechanization, a multipronged approach, will be required. It includes steps to consolidate landholding, innovative rental models; increased R&D and local industries and development of human resource.

- **Conclusion2:**

- Unrestricted mechanization may be associated with certain limitations (unemployment, emissions etc.) and thus is not advisable. India should look into **selective phased mechanization in areas** which are the biggest bottleneck for now.
- Gradually, as the employment situation in other sectors improve and non-renewable energy sector is well-developed, we should move towards complete mechanization of the agri-sector.

2) QUESTION-2

- **Question - 2:**

- Analyze the role of government policies and initiatives in promoting agriculture mechanization in India. Assess the effectiveness of existing schemes and suggest measures to further encourage the adoption of mechanized farming techniques. [15 marks, 250 words]

- **Introduction:** Class Discussion

- The role of government policies and initiatives in promoting agricultural mechanization in India has been significant. The Government has introduced schemes such as the Rashtriya Krishi Vikas Yojana (RKVY), the Sub Mission on Agricultural Mechanization (SMAM), and the Pradhan Mantri Krishi Sampada Yojna. These schemes provide financial support, subsidies, and loans for the purchase of farm machinery, establishment of custom hiring centres, and capacity building activities. Similarly through **FARMS-App** small and marginal farmers are being connected to custom hiring centres.

- Additionally, initiatives like Farm Machinery Training and Testing Institutes (FMTTIs) and Krishi Vigyan Kendras (KVKs) play a crucial role in providing training, demonstration, and technical assistance to farmers.
- **Impact of These Schemes:**
 - **Under SMAM**, as of Dec 2022, more than 21,000 CHCs and more than 18,000 farm machinery banks have been established.
 - **Indian Tractor Industry** is the largest in the world accounting for one-third of the total production.
 - Overall, agri mechanization has increased in the country, which has led to increase in agri productivity, reduction in labor drudgery, and overall positive impact on rural development.
- While, the above positive outcomes have come, many challenges remain. To further, encourage the adoption of mechanized farming techniques, several measures can be considered:
 - **Consolidation of landholding** can go a long way in ensuring the benefits of agricultural mechanization
 - **Enhancement of the availability and affordability of credit** for farmers to purchase machinery.
 - **Innovating a rental model** by institutionalizing for high cost farm machinery such as combine harvester, sugarcane harvester etc. to reduce the cost of operation and it can be adopted by private players or State or Central Organizations in major production hubs.
 - Expanding the network of CHCs by encouraging public private partnerships in setting up and operating CHCs.
 - **Encouraging FPOs** to pool resources, negotiate better prices, and collectively benefit from mechanizations.
 - **Promoting indigenous industry** in the farm equipment sector to reduce the cost and to **promote more R&D in the farm equipment sector** which would contribute to the development of region specific and crop specific machinery.
 - **Public Private Partnership (PPP)** can facilitate technology transfer, provide access to quality machinery, offer maintenance and repair services and ensure availability of spare parts and technical support.
 - **Farmer Education and Training Programs** need to be strengthened. It would enable farmers to better operate the new machinery and contribute to skill development of youth for new kinds of jobs in rural areas.
 - **Policy Reforms:** Streamlining policies on import duties on machinery, taxation, licensing requirements etc. can create a conducive environment for farm mechanization.
- In conclusion, while government policies and initiatives have played a crucial role in promoting agriculture mechanization in India, there is a need for continuous improvement. If the measures suggested above are taken, it would lead to improved agri-productivity, reduced labor drudgery, and overall rural development.

3. CROP INSURANCE

- **Example Questions**
 - i. How is Pradhan Mantri Fasal Bima Yojana an improvement over the previous crop insurance schemes? Discuss the key factors hindering its proper implementation. [15 marks, 250 words]

- ii. What are the key factors which limit the coverage of crop insurance schemes? How is PMFBY trying to solve those problems? [10 marks, 150 words]
- iii. "PMFBY if implemented properly across the country would mitigate farm distress to a large extent especially when the erratic climates have become a norm rather than exception" Elaborate [12.5 marks, 200 words]

- Introduction

- Thousands of farmers end up committing suicide every year. Bad harvest, drought etc. are the main factors, but the absence of any crop insurance forces them to take this extreme step.

- Background: Why didn't Farmers participate in agri-insurance Schemes?

- NSSO report 573 (2012-13) identified following as the main reason for farmers not insuring their crops - **Non awareness, not interested, non-availability, lack of resources, complex process and Delays** in claim payments

- **Problems with Crop Insurance Scheme before PMFBY** -> Partial Risk Coverage; Available only for notified crops; High Premium Rates; Complex system -> lack of uniformity; Delays in claims settlement; very less focus on awareness generation.

- Pradhan Mantri Fasal Beema Yojana (PMFBY1)

▫ Introduction

- In a bid to protect farmers against losses incurred because of frequent changes in weather patterns, the PMFBY was launched in Feb 2016 and was implemented from Kharif 2016 (June 2016). It replaced the NAIS and MNAIS. However, WBCIS and Coconut Palm Insurance Scheme have continued to operate. Premium paid under WBCIS has been brought on par with PMFBY

▫ Key Improvements:

- **Higher losses coverage** (pre harvest to post harvest losses)
 - Provision of claims upto 25% of sum insured for prevented sowing.
 - It covers post-harvest losses also.
 - It expands the definition of disaster to include aspects like flooding of crops and damage after harvest.
 - Provision of individual farm level assessment for Post-harvest losses against the cyclonic & unseasonal rains for the crops kept in the field for drying upto a period of 14 days, throughout the country.
- **Full Coverage:** No upper limit on government subsidy -> Doesn't cap premium rates, so that farmers can get full sum assured.
- **Uniform low premium rates for farmers**

Crop	Premium charged
Kharif	2.0% of sum assured
Rabi	1.5% of sum assured
Annual Commercial and horticulture crops	5% of sum assured

- **Uniformity in implementation:** Districts are allotted to insurance companies on cluster basis for a longer duration to ensure uniformity in implementation of the scheme.
- **Provisions for quick settlement of claims**
 - **Note:** Operational guidelines under PMFBY require state governments to carry out at least four CCEs in every village panchayat for every notified crop and submitted the yield data to insurance companies within a month of the date of harvest. The companies have to settle the claims within three weeks of receiving CCE data.
- **Increased Use of Technology:** The use of technology will be promoted to greatest extent possible. Remote sensing, smart phones and drones will be used for quick estimation of the crop losses and early settlement of claims.
- A provision of Stratified Redressal Mechanism, viz., District Level Grievance Redressal Committee (DGRC), State Level Redressal Committee (SGRC) has been made.
- **The scheme PMFBY and Restructured Weather Based Crop Insurance Scheme were made voluntary for all farmers, post its revamp in Feb 2020.**
- **Technology Interventions in PMFBY:**
 - **National Crop Insurance Portal** has been developed to handle all grievances from end to end. This portal is equipped with the necessary features, such as complaint/Query capturing through multiple modes, farmer authentication etc.
 - **Digi-Claim-Payment-Module:** A new module has been created to integrated National Crop Insurance Program (NCIP) with public financial management system (PFMS) end-to-end. Through this, government will have visibility of the quantum of eligible claims, claims paid by insurance company, and the actual claims transferred to beneficiary farmers.
 - **Yield Estimation Based on Technology (YES-Tech):** It is a technology based yield estimation mechanism which was tested in 100 country districts. Now it is being used by several states.
 - **Weather Information Network & Data System (WINDS):** It is pioneering initiative to set up a network of Automatic Weather System and Rain Gauges at Taluk/Block and Gram Panchayat (GP) levels. It is proposed that an automatic rain gauge (ARG) would be installed at every GP and an Automatic Weather System (AWS) at every block covered by PMFBY.
 - **Collection of Real Time Observation and Photographs of Crops (CROPIC):** In this initiative, periodic photographs of crop are taken in its lifetime. These photographs validate sown and insured crops, assess, crop damage and act as a technology based yield estimation model.
- **The scheme wants to support sustainable production of Agriculture sector by way of financial support; stabilizing income; promoting adoption of innovative and modern practices; ensuring flow of credit to agri-sector.**
- **Progress So far:**
 - As per ESI 2023-24, PMFBY is the largest crop insurance scheme in the world in terms of farmer enrolments, and is the third largest scheme in the world in terms of insurance premium.
 - The overall insured area in 2023-24 reached 610 lakh hectare compared to 500.2 lakh hectare in 2022-23.
 - More than 1.5 lakh crore has been paid under the scheme to farmers.
- **Problems with the scheme/ Why coverage of scheme is going down:**

- **Financial Burden on states** have emerged as a **major issue**.
 - » Many states like Andhra, Telangana, Gujarat, Bihar etc. have bowed out of this scheme
- **Certain losses not covered** - Losses by wild animals, nuclear risks, riots, thefts, enmity etc.
- **Poor Implementation:**
 - » **Delays in Processing of Claims / Non Payment of claim.**
 - The delay is attributed mostly to **problems in Crop Cutting Experiment (CCE)** - Very few CCEs are happening; Use of technology has been minimal.
 - CSE report said that the insurance companies in many cases, **didn't investigate the losses due to a localized calamity** and, therefore, did not pay claims.
 - Insurance companies are not receiving subsidies from center and especially state government fast enough which is further leading to delay in settlement claims.
 - **Grievance redressal mechanism has various flaws:**
 - The Standing Committee on Agriculture was disappointed about no data on complaints resolved under PMFBY (Aug 2021)
- **Fails to cover share croppers and tenant farmers** (just like previous schemes)
- **Insurance companies making huge profits** - High Premium costs compared to insurance claims.
 - » **Actuarial Premium rates are not decreasing with increased coverage** (Gulati's paper)
- **Danger or discouraging mixed cropping and farm diversification**
 - Limited number of crops are notified by states under PMFBY for insurance coverage.
- **ESI 2023-24** also highlighted the issue of cumbersome procedure for paying premium, lack of bank facilities near the village, and poor awareness of scheme among small and marginal farmers are also limiting the impact of the scheme.

- **Way Forward / How to improve the implementations**

- » **Centre must think of taking over the whole subsidy burden**
- » **Promote more agriculture insurance companies** -> competition -> reduction in premium
- » **Ending Delays**
 - **PMFBY timelines** should be strictly adhered to. States will be able to save on the premium cost.
 - State government should submit Yield Data on CCEs to insurance companies on time.
 - State and central governments should pay their share of subsidies on time.
 - Further, Insurance companies should be made liable for penalties in case of delay in payment.
 - **Digitizing land records and linking them with farmer's accounts** will help speed up assessment of the damage to the crop, and in turn, the settlement of the claims.
- » **Sum insured should not be less** than scale of finance and/or cost of production.
- » Efforts to enhance **coverage of tenant and sharecropper farmers** should be enhanced.
- » **Incentivize group of small farmers** and promote group insurance to enhance the coverage
- » **Involve PRIs and Farmers in various stages of implementation**
 - Currently, in notification of the Scheme, notification of the crops and in quickly assessing damages, farmers have no role to play
- » **Develop a robust scheme monitoring and grievance redressal mechanism.**
 - Set up grievance redressal cells in all districts and states.
 - Provide toll free number for information and complaints
 - Time bound resolution of complaints.

4. FERTILIZER AND ASSOCIATED ISSUES

- **Example Questions**

- i. What is Nutrient Based Subsidy scheme? How far has it been successful in achieving its goals? [10 marks, 150 words]
- ii. Discuss the key problems faced by India's Fertilizer Subsidy Program. How has implementation of Direct Benefit Transfer for Fertilizer Subsidy Scheme resolved some of these issues? Will direct cash transfer to farmers be a better option? Give Reasons [15 marks, 250 words]
- iii. Why is there a need to rationalize the consumption of UREA in the country?. Discuss the recent efforts in this direction [10 marks, 150 words]

- **Introduction:**

- » A fertilizer is any organic or inorganic, natural or synthetic material added to soil to supply one or more plant nutrients essentially to the growth of plants.
- » These fertilizers provide **six macro nutrients** and **8 micro-nutrients** to plants for well balanced growth:
 - i. **6 macronutrients:** nitrogen(N), phosphorus(P), potassium(K), Calcium (Ca), magnesium (Mg), and sulphur(S). They are consumed by plants in larger quantities and make the bulk of fertilizers.
 - ii. **8 Micronutrients:** Boron (B), Chlorine (Cl), Copper (Cu), iron (Fe), manganese (Mn), Molybdenum(Mo), Zinc (Zn) and Nickel (Ni).
- » Fertilizer are an important input for agriculture and have played a major role in increasing farm productivity since green revolution.
- » But Indian farmers have often faced difficulties due to shortage of fertilizers in past. So, the government, giving high priority to farmer's welfare, has taken a number of initiatives to ensure supply of fertilizers around the year. Some of these steps which have been taken in recent years are:

1) **FERTILIZER SUBSIDY: KEY ISSUES AND POLICY OPTIONS**

- **Introduction**

- » Fertilizer, along with good quality seeds and water play a critical role in enhancing agriculture productivity. To ensure affordable availability of fertilizers, the Government of India is running two major schemes: Urea Subsidy Scheme and Nutrient Based Subsidy (NBS) Scheme.

- **Nutrient Based Subsidy (NBS) Scheme, 2010**

- » **Background: Product based subsidy**
 - Subsidy burden was increasing drastically.
- » **Key provisions of NBS**
 - Fixed subsidy based on nutrients (both macro and micro)
 - MRP to be fixed by fertilizer companies on the basis of demand and supply but after incorporating the subsidy element.
 - Rate of subsidy is determined by various factors such as international prices, exchange rate, inventory levels etc.

- The NBS scheme currently covers 21 grades of different phosphatic and potassic (P&K) fertilizers including DAP (diammonium phosphate), MOP (Murate of Potash) and other NPK complex fertilizers.
 - **UREA has been kept outside the coverage of the NBS scheme.**
- **Key Aim** -> Reduced Subsidy Burden; increased production of specialized variety of fertilizers; Balanced application; Improved farm output; promote indigenous fertilizer industry
- **Hasn't been as affective** -> Government's subsidy burden still very high -> UREA kept out of NBS, so farmers shifted to UREA -> Balanced Nutrient Goal also missed.
- **UREA Subsidy Scheme**
 - **Introduction:**
 - » To ensure affordable access to fertilizers to farmers, UREA is made available at **statutorily controlled price**, which at present is Rs **5378 per MT** (exclusive of Central/State Tax and other charges towards neem coating).
 - » The difference between the delivered cost of fertilizers at farm gate and MRP payable by farmers is **given as subsidy to the fertilizer manufacturer/importer by Govt.**
 - » It also includes imported urea subsidy which is directed towards import to bridge the gap between assessed demand and indigenous production of Urea in the country. It also includes freight subsidy for movement of across the country.
- **Issue of Diversion**
 - Being super-subsidized, urea is always **prone to diversion for non-agricultural use** - as a binder by plywood/particle board makers, cheap protein source for animal feed manufacturers or adulterant by milk vendors - apart from being smuggled to Nepal and Bangladesh.
- From 2018, the government announced the **implementation of DBT** for disbursement of fertilizer subsidy.
 - Now the subsidy transfer only happens after the actual sales to farmers by retailers. Retailers have a point of sale (PoS) machine linked to e-Urvarak DBT Portal. Fertilizer buyers (farmers) are required to furnish Aadhar or KCC number.
 - **Advantages**
 - » **Prevents diversion and plug the leakages** (because Aadhar is used)
 - » Timely payment of Urea subsidy to urea manufacturing companies.
 - » Adequate availability of UREA to farmers at adequate prices.
- **Limitations**
 - PoS devices at fertilizer outlets are not integrated with land record data.
 - Any person having an Aadhaar, whether a farmer or not, can buy any quantity of fertilizers.
 - No limit on sale of fertilizers to one person or one family.
 - Thus diversion has continued.
- **Impact of Fertilizer Subsidy Policies in India**

- **Positive Impact:** Fertilizer subsidy policies have had positive impact in terms of **increasing fertilizer consumption** and hence leading to an increase in overall Agri-production.
- **But it has failed in the goals of increasing domestic production (and has thus increased import dependency) and promoting balanced use of nutrients by farmers.** It has also led to diversion of UREA for other industries and exports to neighboring countries.
- **Why domestic industry couldn't be promoted:**
 - **Lack of raw material for potassic and phosphatic fertilizers:** India completely lacks any commercially exploitable source of Potash and the entire demand for MOP is met through import. In the Phosphatic sector (for DAP etc.), there is limited availability of raw material like Sulphur and rock phosphates and hence, a bulk of raw material is imported. Even the **Urea sector** is dependent on imported fuel sources like crude oil and now, even gas.
 - **Low investment in fertilizer sector** over the years.
 - **Inefficient Fertilizer manufacturer companies** - Since they get subsidy based on the cost of production (rather than fixed subsidy for all manufacturers on the basis of per unit production).
- **Increased dependency on Urea has been harmful**
 - **The Lok Sabha Standing Committee on Chemical Fertilizers**, in their Thirty Ninth report, titled '**Nano-Fertilizers for sustainable crop production and maintaining soil health**' highlighted the following issues:
 - i. **Fertilizer Consumption in India** is imbalanced, and Urea accounts for more than 82% of the nitrogenous fertilizers applied to the majority of crops.
 - ii. **NPK Ratio has widened** from 4:3.2:1 in 2009-10 to **7:2.8:1** in 2019-20.
 - Note: The ideal ratio of NPK fertilizer use is considered as 4:2:1.
 - Widespread deficiency of secondary and micro-nutrients. (For e.g. deficiency of zinc is causing the problem of stunting in India)
 - Declining fertilizer response and efficiency due to imbalanced use of nutrients.
 - **Environmental and Health Damage:** Excess nitrogen is leading to nitrogen pollution causing air, water, and soil pollution. Part of Nitrogen leaches down as NO_3 and contaminates the ground water resources. It has been found to cause methemoglobinemia or the Blue baby syndrome.
 - Hampers crop diversification: As farmers tend to stick to those crops where UREA can be used as fuel. Specialized fruits, vegetables require special non-Urea Fertilizers.
- **Diversion of UREA** -> smuggling to neighboring Nepal, Diversion to Industries
- **Huge Fiscal Burden on Government**

Some steps that have been taken to make UREA sector efficient

- i. **GAS Price Pooling**
 - Earlier, different urea plants got gas at different prices, so their cost of production differed.

- Therefore, in 2015 government has approved a major policy intervention. Under this policy the **domestic gas is pooled with imported LNG gas** to provide uniform natural gas to all the Urea manufacturing plants for the production of Urea.
 - **Cost of UREA at pooled price will be less than the price of imported urea.** This will help in increasing the production. This will augment indigenous production capacity.
- ii. **Neem Coating of UREA**
 - Reduces rate of dissolution in soil -> slowly absorbed by plants
 - Reduces diversion to industry
 - Neem has other advantages for crops -> insecticidal and pesticidal properties
 - UREA can't be used in synthetic milk now
- iii. **New Urea Policy** to increase the productivity, efficiency and indigenous production
- iv. **Introduction of 45 kg Urea Bag** (from earlier 50 kg) -> aimed at cutting demand
- vi. **Nano Urea and Nano-DAP**
 - Government has notified the specification of Nano nitrogen under Fertilizer Control Order, 1985.
- vii. **One Nation One Fertilizer Scheme**
 - It aims to ensure timely supply of fertilizers as well as eliminate the dilemma of farmers in choosing one of the many brands available in the market.
- viii. **Pradhan Mantri Krishi Samridhhi Kendra (PMKSK)**
 - It has been decided to convert the existing village/block/sub-district/taluk and district level fertilizer retail shop into Model Fertilizer Retail Shops. These shops will act as "One Stop Shop" for all the agriculture related inputs and services.
- ix. **PM PRANAM (Proposed)**
 - Aimed at reducing the use of chemical fertilizers and thus reducing the subsidy burden.
- **Steps that further needs to be taken/Way forward**
 - i. **Work towards self-reliance and reducing dependency on imports**
 - This will protect us from high volatility of fertilizer prices in international market.
 - Government is already working towards this and urea plants at Gorakhpur, Sindri, Brauni, Talcher etc.
 - **Tie-up with gulf countries to set up plants**
 - Gas prices for UREA plants in India is 3 times higher than in gulf countries. So we should set up joint ventures in countries where these prices are low.
 - ii. **Correcting Price Signals and Decontrolling the Fertilizer sector -> More investment and more competition**
 - The Commission on Agriculture costs and Prices (CACP) has recommended the centre to bring urea under NBS regime.
 - iii. **Using Agri-Stack and e-Rupi to improve targeting of fertilizer subsidy** - Limit the amount of subsidized fertilizer based on the size of land and sown crop.
 - iv. **Improve fertilizer efficiency -> Expand the soil health testing facilities and Issue of soil health cards**
 - v. **Digitization of land records**
 - The process of was launched in 2008, but has not gathered momentum.

- Without setting right the land records, it will be impossible to transfer the subsidy to beneficiaries or to issue soil health cards.
- vi. **Ensuring timely reach of subsidy to farmers**
 - Last, but not the least, in the drive for increased efficiency and productivity, we should not forget the question of equity and inclusiveness, for 85% of our operational holdings belong to small and marginal farmers and smaller farmers tend to use fertilizers more intensely.
- vii. **Develop Alternative sources of nutrition for agriculture** - Shift towards non-chemical form of fertilizers -> scope to use large biomass of plants which is wasted today.
- **Conclusion:**
 - » These steps will go a long way in enhancing the productivity of agriculture, mitigating climate change, providing an alternative to chemical fertilizers and balancing the fiscal impact of fertilizer subsidy on the Union Budgets in the years to come.

2) USING AGRI-STACK TO IMPROVE THE TARGETING OF FERTILIZER SUBSIDY (ESI 2023-24)

- **Practice Question:**
 - » Discuss how Agri-Stack can contribute to rationalizing the farm subsidy situation in India [10 marks, 150 words]
- **AgriStack** is the digital foundation set up by government of India to make it easier to bring various stakeholders together to improve agriculture in India and enable better outcomes and results for farmers by using data and digital service. It is fairly well developed in several Indian states and can provide the right tool through which fertilizer subsidy may be better targeted.
- It will ensure that subsidized fertilizers are sold only to those identified as farmers or authorized by farmers.
 - » Quantity of subsidized fertilizer is fixed based on parameters such as land ownership and prominent crop of the district. The parameters may be later refined based on crop grown and soil nutrient status (in convergence with SHC scheme).
 - » Provision for top-up entitlement can be provided in case of crop damage or calamities caused by volatilities in weather conditions, in convergence with State Disaster Response Fund and National Disaster Response Fund.
 - » E-RUPI, a seamless one time payment mechanism, can be utilized to provide the necessary subsidy to the farmer directly.
 - » If a farmer purchases a quantity of fertilizers that is less than their entitlement, in that case the remaining subsidy can be used to purchase other agricultural input, such as seeds and pesticides, also sold at these outlets.
- **Advantages:**

- Streamlines subsidy distribution process;
 - Prevents diversion for non-agri purposes
 - Incentivizes farmers not to use UREA excessively on account of being cheaper than NPK fertilizer and thus can lead to balanced use of fertilizer as per the requirements of crop and soil.
- **Way Forward:**
- **Integration of PoS devices with farmer's registry** in Agri Stack and farmer's registry will include the Aadhaar number of each farmer, details of all agricultural lands owned by the farmers, and dynamic Updation of land ownership data through mutation module.
 - **Facility to put details** of other authorized buyers
 - **Facility to update bank details, mobile numbers and other details of farmers**
 - **Run a pilot project** in one district of each state and based on the outcomes of these parallels, the decision of the future mode of fertilizers subsidy administration may be made.

5. MAKING AGRICULTURE SUSTAINABLE

- **Agriculture sector** is facing several sustainability challenges:
 - a. **Deforestation**: Over the decades, the need to increase agriculture production has led to destruction of forests and biodiversity dependent on them.
 - b. **Inefficiency and Resource wastage**: For e.g. 80% of water use in India goes for agriculture and in flood irrigation, around 70% of the water is wasted.
 - c. **Excessive use of fertilizers, insecticide, pesticide etc:**
 - While, the per hectare usage of agriculture chemicals in India remains significantly lower than in most developed countries, its use of chemical fertilizers have increased over the years.
 - The **Urea Subsidy Scheme** has led to excess use of urea contributing to imbalance in the major plant nutrients including Nitrogen-Phosphorus-Potassium (NPK).
 - This imbalance is harming the efficiency of fertilizers; soil quality and thus overall productivity.
 - It is also leading to environmental damages through eutrophication and creation of dead zones in water bodies.
 - d. **Air Pollution and Water Pollution**: Practices like stubble burning are contributing to the problem of air pollution and excessive use of urea is leading to water pollution.
 - e. **Land Degradation**: Other than overuse of chemical fertilize, insecticide etc, it is also being caused by excessive irrigation.
 - f. **Climate Change**:
 - **Agriculture has led to climate change in several ways** -> Deforestation; methane emission by rice cultivation; production of fertilizers use fossil fuels; electricity subsidy leads to excess use of electricity which is mostly coming from fossil fuel.
 - **Agriculture is also getting negatively impacted by climate change:**

- **Variability in weather conditions and the relative pre-dominance of rainfed agriculture** also impact production and productivity. As per ESI 2023-24, if appropriate measures are not taken, rainfed rice yield in India are projected to drop by 20% in 2050 and 47% by 2080; irrigated rice yield is also expected to drop by 5% in 2080 scenario.
 - **Further, higher temperature** itself may create several challenges for crops growth.
 - g. **Anti-biotic resistance:** Excess use of anti-biotics in fruits, vegetables, poultry and animal husbandry sector is increasing the scope of anti-biotic resistance.
 - h. **Land Fragmentation:** With increasing population there is an increasing tendency towards land fragmentation which is hampering long term investment, mechanization and thus hampering sustainability of agriculture.
- **Key steps taken:**
- a. **National Mission for Sustainable Agriculture (NMSA)**, a part of the NAPCC, aims to evolve and implement strategies to make Indian agriculture more resilient to the changing climate.
 - b. **For Assuring Irrigation for farmlands**, several steps have been taken:
 - Rainfed Area Development (RAD), being implemented under NMSA, adopts an area based approach for development and conservation of natural resources through watershed development and soil conservation activities.
 - Pradhan Mantri Krishi Sinchayi Yojna (PMKSY) consisting of two major components - Accelerated Irrigation Benefit Program and Har Khet ko Pani (HKKP), promote the extension of area under irrigation and water efficiency.
 - Impact:
 - Irrigation Area Coverage (IAC) increased from 49.3% of gross capped area (GCA) to 55% in 2020-21.
 - Irrigation Intensity (Ratio of gross irrigated area to net irrigated area) recorded an increase from 144.2% in 2015-16 to around 154.5% in 2020-21.
 - PDMC Scheme also support micro-level water harvesting, storage, management, etc.
 - Impact: An area of 90.0 lakh hectares has been covered under micro-irrigation in the country under the PDMC from 2015-16 to 2023-24.
 - A Micro-Irrigation Fund (MIF) of an initial corpus of Rs 5,000 crore has also been created with NABARD to facilitate the states in mobilising the resources for expanding the coverage of micro-irrigation.
 - c. **Efforts towards reducing the use of chemical fertilizers:**
 - Soil health Card (SHC) initiative for promoting balanced application of fertilizers
 - PM Programme for Restoration, Awareness Generation, Nourishment, and Amelioration of Mother Earth (PM-PRANAM) incentivizes states to reduce the chemical fertilizer use. It promotes sustainable methods such as the use of alternative fertilizers, Viz., Nano-Urea, Nano-DAP, and Organic Fertilizers.
 - Under the scheme, 50% of the fertilizer subsidy saved by the states in particular financial year will be passed on to the states/UT as a grant.
 - The introduction of "Urea Gold", which is urea infused with sulphur to address sulphur deficiencies, has been another measure to improve the nutrient balance in the soil.
 - d. **Efforts to promote organic and natural farming:**

- Paramparagat Krishi Vikas Yojna (PKVY) and Mission Organic Value Chain Development for North East Region (MOVCDNER) are two schemes running since 2015 promoting organic farming.
- e. **Reducing Stubble Burning:** The Crop Residue Management Scheme from 2018-19 to support the efforts of the Punjab, Haryana, Uttar Pradesh and NCT of Delhi to address air pollution and subsidize machinery required to manage crop residue.
 - Under the scheme, financial support is also given to take up large-scale demonstration of bio-decomposer on farmer's fields that accelerates the in-situ decomposition of paddy straw.
- f. **Promoting mechanization** to reduce the need of stubble burning
- g. **Steps to promote integrated farming; mixed agriculture; vertical farming etc.**

6. ORGANIC FARMING

1) QUESTION-1:

Discuss the concept of Organic Farming and its significance in promoting sustainable agriculture. Examine the benefits of Organic Farming in India. (15 marks)

- **Introduction:**

» **Definition based:**

- Organic farming revolves around the idea of cultivating crops and raising livestock with natural inputs without using chemical products like chemical fertilizers, insecticides, pesticides etc.

» **Data based:**

- As per ESI 2023-24, India has 68.05 lakh hectare under organic farming in 2022-23. Sikkim is the first state of India (and in world) to become fully organic. States including Tripura and Uttarakhand have also set similar targets.

- **Key Features/Practices:**

- » It prohibits (with rare exceptions) synthetic fertilizers, pesticides, insecticides, herbicides, GMO, antibiotics and growth hormones and relies on on-farm inputs such as crop residues, farmyard manure, enriched composts, vermi-compost, oil cakes, bio-fertilizers etc. for nutrient management of crops.
- » Pests and diseases are managed by eco-friendly practices of crop rotation, trap crops, bio-pesticides like neem-based formulations, bio control agents, mechanical traps, stale seed bed etc.

- **Advantages:** The Centre for Science and Environment (CSE) has in its report, "Evidence (2004-20) on Holistic Benefits of Organic and natural farming in India" consolidated long-term evidence on holistic benefits of organic and natural farming:

- » **Higher Yield:** Long term trend on crop yield shows that organic approach is better than inorganic one.
- » **Cost of organic cultivation** though higher right now, can reduce if the farmer starts producing and using organic and bio-inputs herself.
- » **Promotes Sustainability in Agriculture:**

- **Maintain long term soil fertility** by optimizing conditions for biological activity within the soil. Organic approach is better at ensuring macro and micro nutrients, organic carbon and rhizosphere microbiome.
- **Recycling of Resources** -> Reduced cost of production
- Relies on renewable resource in locally organized agriculture sector.
- Provides alternative care that promotes the health and meets the behaviours needs of the livestock.
- » **Food Quality and Nutrition** with organic approach are found to be better than inorganic. Evidence shows that quality parameters like carotenoids, total soluble solids, vitamin C, total sugars, and lycopene are higher in organic approach in vegetables.
- » **Protects Environment** - reduces soil pollution, water pollution etc.
- » **Mitigates climate change** - by reducing dependence on industry produced synthetic chemicals.
- **Conclusion:**
 - » Overall, Organic Farming in India aligns with the principles of sustainable agriculture, fostering ecological balance, farmer livelihoods, and consumer health. Its emphasis on environmental stewardship, soil health, and biodiversity conservation makes it a significant approach to address the challenges of modern agriculture and promote a sustainable and resilient food system

2) QUESTION-2

- Evaluate the role of government policies and initiatives in promoting Organic Farming in India. Assess the effectiveness of existing schemes and suggest measures to further encourage the adoption of Organic Farming practices [15 marks, 150 words]
- **Introduction:** Organic farming in India aligns with the principles of sustainable agriculture, fostering ecological balance, farmers livelihoods, and consumer health. Keeping this significance in mind, the Indian government has implemented several policies and initiatives to promote organic farming in the country:
- **Key Initiatives include:**
 - » **Financial Support for Transition:** Through initiatives like Paramparagat Krishi Vikas Yojna, Bhartiya Prakritik Krishi Padhati Yojana, Mission Organic Value Chain Development for North-East Region (MOVCD - NER), government provides financial support for organic inputs, capacity building, certification and infrastructure development.
 - » **Certification Support:**
 - **National Program for Organic Product (NPOP)** provides certification mechanism for primarily export products.
 - **Participatory Guarantee System for India (PGS-India)** by MoA&FW certifies cluster of small farmers.
 - » **Market Linkages:**
 - Through initiatives like Jaivik Kheti Portal, which serves as an e-commerce cum knowledge platform, government help in connecting organic farmers with consumers, retailers and exporters.

- » **Capacity Building:** Through institutions like Agricultural and Processed Food Products Export Development Authority (APEDA), National Centre of Organic Farming (NCOF), and Krishi Vigyan Kendras (KVKs), government supports capacity building programs which contribute to training, demonstration, and technical assistance to farmers on organic farming practices.
- While government policies and initiatives have contributed to the growth of organic farming in India, certain challenges persist:
 - » **Limited Awareness and outreach**
 - » **Certification is expensive** under NPOP and is out of reach for most of small and marginal farmers.
 - » **Insufficient infrastructure development** for organic processing, storage, and value chain
 - » **Market** for organic products is either in Urban areas or in export
 - Accessing such markets require contracts with larger companies
 - A small farmer can't reach those who may pay more for organic produce
 - » **Other Factors hindering growth of organic farming in India**
 - Lack of conviction among policy makers
 - Limited consensus among scientific communities in favor of organic and natural farming
 - Tendency to evaluate the non-chemical agricultural practices only on the basis of yield.
- **Way forward to reduce the above shortcomings limitations**
 - » **Existing Programs** to support organic farming needs to be **scaled up, expanded and properly funded**.
 - » **Strengthen Extension Services** to provide farmers with information, training, and technical support on organic farming practices.
 - » **Government should draw a roadmap** that sets the long-term agenda for adoption of agro-ecological approaches across different parts of the country.
 - This roadmap should also consider mechanisms to incentivize farmers to adopt agro-ecological practices such as payments for ecosystem services.
 - » Measures to adequately produce and make available **quality organic fertilizers and biofertilizers at low cost** should be priority.
 - » **Simplified Certification Mechanisms:**
 - The organic certification needs to be made farmer friendly and low cost. Measures should be taken to address the problems of PGS-India (Participatory Guarantee System of India) certification system and its implementation.
 - An alternative certification that is simpler for farmers and trustworthy for consumers could be explored for well-connected local markets.
 - » **Problems of small and marginal farmers** can be reduced by bringing them together for collective farming.
 - Government should also ensure that farmers organize in a way they can enter into contracts and demand a fair price from global companies.
 - » **Enhance R&D** specific to organic farming for the development of region specific organic farming techniques, pest and disease management strategies, and organic input formulations.

3) ORGANIC FARMING VS NATURAL FARMING

- Compare and contrast Organic Farming and Natural Farming. Discuss their principles, practices and potential benefits of each approach in promoting sustainable agriculture [15 marks, 250 words]
- **Introduction:** Organic Farming and Natural Farming are two approaches to agriculture that share a common goal of promoting sustainable and environmentally friendly farming practices. However, they differ in their principles, practices, and approaches. Let's compare and contrast Organic Farming and Natural Farming in terms of their principles, practices, and potential benefits in promoting sustainable agriculture:
- **Principles:**
 - **Organic Farming:** It is based on the principle of maintaining soil health, ecological balance, and biodiversity. It focuses on the use of natural inputs, such as organic manures, compost, and biofertilizers, while prohibiting the use of synthetic chemicals, GMOs, and irradiation.
 - **Natural Farming:** It is based on the principle of working in harmony with nature and minimizing external interventions. It emphasizes the use of locally available resources and indigenous microorganisms to nourish the soil and enhance plant growth, aiming to restore the natural balance of the farming system.
- **Practices:**
 - **Organic Farming:** It practices include crop rotation, organic fertilization, composting, vermiculture, biological pest control, and weed management. It involves using approved organic inputs, such as organic manures, green manures, and organic-approved biopesticides.
 - **Natural Farming:** It practices focus on creating on-farm resources and reducing external inputs. It involves the use of indigenous microorganisms (IMO), fermented solutions, herbal extracts, and other naturally available substances to improve soil fertility and control pests and diseases. Natural Farming discourages the use of external inputs, including organic fertilizers and pesticides.
- **Potential Benefits in Promoting Sustainable Agriculture:**
 1. **Organic Farming:**
 - Improves soil health and fertility by promoting organic matter content and microbial activity.
 - Reduces environmental pollution and minimizes the negative impacts of synthetic chemicals on ecosystems.
 - Enhances biodiversity by preserving natural habitats, supporting beneficial insects, and protecting pollinators.
 - Produces nutritious and safe food with reduced pesticide residues and chemical additives.
 - Supports rural development by creating opportunities for organic markets and value-added products.
 2. **Natural Farming:**
 - Promotes soil regeneration and improves soil structure and fertility.
 - Reduces dependence on external inputs and chemical-based fertilizers, leading to cost savings for farmers.

- Enhances the resilience of crops to pests and diseases through the use of indigenous microorganisms.
- Conserves water resources by promoting efficient water management practices.
- Supports self-sufficiency and sustainable livelihoods for farmers by relying on locally available resources.

▪ **Conclusion:**

- Both Organic Farming and Natural Farming contribute to sustainable agriculture by minimizing environmental impact, promoting soil health, and producing safe and nutritious food. They emphasize a holistic approach to farming that considers the long-term ecological balance and the well-being of farmers and consumers. The choice between the two approaches depends on factors such as local conditions, available resources, and farmer preferences

7. ZERO BUDGET NATURAL FARMING (ZBNF)

- **Question1:** Explain the concept of Zero Budget Natural Farming (ZBNF) and its significance in promoting sustainable agriculture. Discuss the principles, practices, and potential benefits of ZBNF in India. [15 marks, 250 words]
- **Introduction: Background/Context:** The **neo liberalisation** of Indian economy has led to a **deep agrarian crisis** which has led to small scale farming becoming an unviable vocation. This is due to high cost of inputs, increased dependency on debt, volatile markets, environment pollution etc. Here, ZBNF has emerged as a ray of hope.
- **Definition: What is ZBNF?**
 - ZBNF is a method of **chemical-free agriculture drawing from traditional Indian practices**. It is a set of agricultural methods which doesn't involve any credit, doesn't spend any money on purchased inputs (zero budget), uses very less water (10%), doesn't use any synthetic chemical fertilizer or pesticide (thus natural) and believes in natural growth of crops using inputs which are locally available.
- **Subhash Palekar**, the original promoter of ZBNF, **identifies four important Pillars of ZBNF.**
 - a. **Beejamrita/beejamrutha** is the **seed treatment using local cow dung, cow urine, lime and soil**. This protects young roots from fungus and soil borne or seed borne diseases.
 - b. **Jivamrita/Jivamrutha** is a fermented microbial culture which is used as a base fertilizer. It is a fermented mixture of water, cow dung, cow urine, jaggery, pulse flour and a handful of soil from the bund of the farm. It provides basic nutrient, acts as catalytic agent to promote the activity of micro-organisms, and reduces fungal and bacterial infections.
 - c. **Acchadana - Mulching** activities to ensure favourable microclimate in the soil. Three kinds of mulching has been suggested: **Soil Mulch**, **Straw Mulch** and **Live Mulch** (symbiotic intercrops and mixed crops)
 - d. **Whapasa - Moisture:**
 - Palekar questions the idea that plant roots need a lot of water and therefore criticizes over reliance on irrigation in green revolution. According to Palekar, root needs water vapours.



- **Whapasa is a condition where there is both air molecule and water molecule present in the soil** and thus Palekar encourages reducing irrigation, irrigating only at noon, and in alternate furrows.
- **In addition**, ZBNF includes three methods of insect and pest management: Agniastra, Brahmastra, and Neemastra (all different preparation using cow urine, cow dung, tobacco, fruits, green chilli, garlic and neem).
- **Benefits of ZBNF**
 - ZBNF has not only worked in agronomic terms. But, it has brought about a variety of social and economic benefits.
 - **Reduced Cost: Reduces Resource Utilization -> Higher income for farmers**
 - For e.g. a study from Andhra Pradesh found that ZBNF in paddy farming has led to a saving of 1,400 to 3,500 cubic meters of water per acre per paddy cropping period. This was achieved by increasing the time interval of irrigation cycles
 - **Sustainability:**
 - **Seed Diversity:** Rather than standard GM crops, it promotes local variety of seeds, their conservation and reuse.
 - **Water Conservation**
 - **Biodiversity Conservation:** ZBNF encourages the planting of diverse crops, inter cropping, and maintaining natural habitats on the farm. This promotes biodiversity, preserves native species, and supports the ecosystem balance.
 - **Reduced Pollution and GHG emissions:** By avoiding synthetic chemicals in agriculture, ZBNF promotes soil conservation, reduces water pollution and also controls global warming.
 - **Food Safety** due to less chemical use
 - **Household Food Autonomy**, not dependent on MNCs for seeds and fertilizers
 - **Reduced Import Dependence** as India is a net importer of fertilizers
 - **Reduced Subsidy burden for government** -> More spending on other socio-economic sectors
- **Some Limitations**
 - **Critics of ZBNF**, including some experts from NITI Aayog note that India needed the Green Revolution in order to become self-sufficient and ensure food security. Therefore they **warn against a wholesale move away from the model without sufficient proof that yields will not be affected**.
 - Further, some experts feel that **Mr. Palekar's claims stand agriculture science on its head**. Indian soils are poor in organic matter content. About 59% of soils are low in available nitrogen; about 49% are low in available phosphorus; and about 48% are low or medium in available potassium. Indian soils are also varyingly deficient in micronutrients, such as zinc, iron, manganese, copper, molybdenum, and boron.
 - They feel that Mr Palekar has a **irrational position on the nutrient requirements of plants**. According to him, 98.5% of the nutrients that plants require is obtained from air, water and sunlight; only 1.5% is from the soil. All nutrients are present in adequate quantity in all types of soil. Jivamrita make these nutrients available to the plants by increasing the population of soil micro-organisms. Many experts consider **all these as baseless claims**. Prescribed Jivamrita will only provide 750 g of nitrogen per acre per

season which will be **inadequate considering the nitrogen requirements of the Indian Soils.**

- Several reports talk about farmers in Maharashtra returning back to older methods as ZBNF returns drop after a few years.
- Further, Sikkim, which has seen some decline in yields following a conversion to organic farming, is used as a cautionary tale regarding the pitfalls of abandoning chemical fertilizers.

- **Way forward**

- **Multi-location studies** are needed to be scientifically conducted to validate long term impact and viability of the model before it can be scaled and promoted country-wide.
- If **found successful**, an enabling institutional mechanism could be setup to promote the technology. At the same time, if it turns out to be unsuccessful and unscientific, its inclusion in our agri-policy will be unwise and imprudent.

- **Conclusion**

- ZBNF seems to be a possible alternative to solve the issue of soil degradation, water pollution and to promote sustainable agriculture. But, it should be formally adopted and expanded once it is backed by scientific studies and analysis.

8. INTEGRATED FARMING SYSTEM

1) QUESTION-1:

- How far is Integrated Farming System (IFS) helpful in sustaining agricultural production. [10 marks, 150 words] [CSE Mains 2019]
- **Introduction** Integrated Farming System (IFS) is a comprehensive farming approach that combines multiple agricultural activities and components within farming system to optimize resource utilization, increase productivity, and improve sustainability. It involves the integration of crops, livestock, fisheries, agroforestry, and other allied enterprises (renewable energy - biogas generation) in a synergistic manner.
- **IFS** is helpful in sustaining agricultural production in the following ways:
 - i. **Resource Optimization and Enhanced Productivity:** Integration of various components like crops and livestock can make efficient use of resources and minimize waste.
 - **For example**, crop residues and agricultural by-products can be utilized as livestock feed, while animal waste can be recycled as organic fertilizer for crops
 - ii. **Nutrient Cycling and Soil health:**
 - Livestock manure and crop residues serve as organic fertilizers, improving soil fertility and nutrient availability.
 - The integration of leguminous crops in crop rotations adds nitrogen to the soil through biological nitrogen fixation.
 - Soil Conservation Practices like contour ploughing and agroforestry, help prevent erosion and maintain soil health.
 - iii. **Pest and Disease management:**
 - Diversity of crops and livestock reduces the risk of pest and disease outbreaks.

- Crop rotation, inter-cropping, and mixed cropping help disrupt pest life cycles and reduce pest pressure.
- iv. **Climate Resilience:**
 - The integration of trees and agro-forestry practices help mitigate climate risks by providing shade, reducing wind speed, improving water filtration, and sequestering carbon.
 - Diverse Crop pattern is also more adaptable to climate variability.
- v. **Economic security through income diversification:** IFS offers multiple income stream (crops, livestock, etc.) which reduces income risk associated with single crop.
- vi. **Environmental Sustainability:**
 - **Reduced Pollution and land degradation:** IFS reduces the need for synthetic fertilizers, mitigates nutrient runoff and pollution, and enhances soil health and long-term sustainability
 - **Biodiversity Protection:** The integration of diverse components helps conserve biodiversity, protect natural habitats, and provide ecological niche for beneficial organisms.
- **Conclusion:**
 - Keeping these advantages in mind Indian Council of Agriculture Research along with various central and state agricultural universities have developed many different IFS models suitable for different geographical regions of the country. These models are being disseminated to small and marginal farmers through the nation-wide chains of Krishi Vigyan Kendras.

2) QUESTION-2:

- Discuss the challenges and constraints faced by farmers in adopting Integrated Farming Systems and suggest strategies to overcome them [15 marks, 150 words]
- **Introduction:**
 - Integrated Farming System (IFS) is a more sustainable and profitable way of agriculture. In spite of this, its adoption is very less due to several factors hindering its adoption.
- **The Key Challenges and Constraints** that farmers face when adopting IFS include:
 - **Land Constraints and Fragmentation:** Small and marginal farmers may find it difficult to implement various components of IFS.
 - **High initial investment** in things like new equipment, training, and infrastructure. This can be a barrier for small-scale farmers.
 - **Lack of knowledge and skill** can be a challenge, as farmers need to be able to understand the principles of IFS and how to implement them on their farms.
 - **Limited access to markets** can discourage farmers from adopting IFS, as they may not see the financial benefits of doing so.
 - **Government policies** like MSP regime, Urea subsidy etc. may prevent farmers from diversifying more sustainable practices.

- **Cultural and Social Factors:** Traditional Farming Practices, social norms, and cultural beliefs can pose challenges to the adoption
- **Way Forward:**
 - **Land Consolidation Arrangements,** Land Leasing Reforms, and land pooling can enable farmers to access larger land area for implementing integrated farming practices.
 - **Work on access to affordable credit, subsidies** to help farmer overcome financial constraints. Reforming agri-credit system can be crucial here.
 - **Increased availability of affordable quality seeds, livestock breeds, fertilizers, feeds** etc. would be very crucial for implementing IFS.
 - **Training and Extension Programs** should be conducted to educate farmers about the IFS. Agri-universities, KVKs, and NGOs can play a crucial role in disseminating information and providing technical support.
 - **Awareness Campaigns, Social Mobilization, and engagement** with local communities can help overcome the cultural and social hindrances to IFS.
 - **More Research and Development (R&D)** to develop unique sustainable IFS system for different geographical regions.
- **Conclusion:**
 - Collaboration among farmers, research institutions, government agencies, and civil society organizations is crucial to provide the necessary guidance, resources, and support for farmers to successfully adopt and implement Integrated Farming Systems

9. PRECISION AGRICULTURE

- **Why in news?**
 - A session on “**Sensors and Sensing for Precision Agriculture**” was organized under “**Precision Agriculture**” by ICAR-Indian Agricultural Research Institute on 05th October 2020, as **part of the Vaishwik Bhartiya Vaigyanik (VAIBHAV) Summit 2020**.
 - This is a **Government of India initiative** to bring together the thought process, practices, R&D culture of Overseas and Indian scientists/academicians through a series of structured deliberations and constructive dialogue, and develop a road map for translational research/academic culture for tangible output and strengthening the S&T base for providing the impetus to endeavor of Aatma Nirbhar Bharat (Oct 2020)
- **Example Question**
 - To feed 1.5 billion people by 2030, it is imperative to accelerate adoption of precision agriculture technologies" Elaborate [10 marks, 150 words]
- **About Precision Agriculture**
 - Precision Agriculture is a technique of agriculture which uses technology to determine the exact amount of input (water, fertilizers, pesticides etc.) required to ensure crops and soil receive exactly what they need for optimum health and productivity.

- This kind of agriculture is highly dependent on technology - Specialized equipment, software and IT.
 - It requires accessing real time data about the conditions of crops, soil and ambient air. It also needs hyper local weather predictions, labor costs, and equipment availability.
 - **Sensors** in the field measure the moisture content and temperature of the soil and surrounding air.
 - **Satellites and Robotic drones** provide real time images of individual farmers.
- **Advantages**
 - **Economic Benefits**
 - Increase agri-productivity
 - Improve the quality and reduce the cost of production
 - Improved socio-economic condition of farmers
 - **Improve Sustainability of Agriculture**
 - Prevent Soil Degradation
 - Reduce Chemical application
 - Efficient resource use (water, fertilizers etc.)
 - E.g. Drip Irrigation used with PA technology can reduce the amount of water used in crops.
 - **PA enables Climate-Smart Agriculture**
- **Challenges**
 - **Small and Marginal Land Holding** in India
 - **Educational challenges** - lack of technical expertise, knowledge and technology
 - **Economic Viability** - High Initial Costs
 - **Heterogeneity of cropping system** and market imperfections.
- **Way Forward**
 - **Identify the niche areas** where crop specific PA system can be promoted.
 - **Creation of a multi-disciplinary team** involving agri-scientists, engineers, manufacturers, economists and farmers to study the overall scope of precision agriculture.
 - Provide **support to farmers** to develop pilots or models, which can be replicated on a large scale.
 - **Creating awareness among farmers** of the consequence of over application of water, chemicals etc. and how PA can be a solution.
 - **Promoting Agri-tech industry** in India by increasing engagement with accelerators, venture capital firms, impact investors etc.

10. MULTI-LAYERED FARMING (MLF)

- **Question:** Discuss the concept of multi-layered farming and its significance in the context of sustainable agriculture in India [10 marks, 150 words]
- It comprises of growing compatible plants of different heights on the same field at the same time. It is mostly practiced in orchards and plantation crops for maximum use of solar energy even under high planting density.
- **Advantages**
 - **Efficient Land Use**

- **Diversification and Risk Reduction**
 - By growing variety of crops, farmers can diversify their produce, accessing different markets, and reducing price risks.
 - **Better Income:** The sale of high value crops, such as fruits, vegetables, herbs, and spices can generate higher returns compared to traditional mono-cropping.
 - **Resource Efficiency:**
 - For e.g. different layers of crops create micro-climates that reduce water evaporation and help retain soil moisture.
 - The **crops complement each other** - for e.g. by providing shade canopy, litter, increasing moisture holding capacity of soil while nurturing microflora.
 - **Nutrient Cycling and Soil Health:** it promotes the recycling of nutrients within the system, enhancing soil fertility and reducing the need of synthetic fertilizers
 - **Biodiversity Conservation:** MLF creates diverse and complex habitats, supporting a range of beneficial organisms, including pollinators, beneficial insects and natural predators. Multiple crops and flowering plants provide food and shelter for a variety of beneficial organisms.
 - **Income generation and livelihood improvement:**
- **E.g.**
 - **Coconut based multilayered farming** (Coconut, pepper, nutmeg, banana, cinnamon, turmeric, ginger) is more remunerative than traditional systems.
 - **Steps being taken in India to promote Multi-Layered Farming**
 - The Indian Institute of Farming Systems Research, Modipuram, Meerut, is **undertaking research (on-station) and technology validation** through farmer's participatory research (on-farm research) on Integrated Farming Systems and Cropping systems in 24 states.
 - **Multi-layer Farming Models on high-value vegetable cultivation** under a three-tier system was introduced in the backward districts of Bihar by ICAR and farmers were able to grow three different vegetables on the same piece of land at a time.
 - **Way Forward:** Class Discussion

11. VERTICAL FARMING

- **Introduction:** Vertical farming refers to the system of cultivating crops in vertically stacked layers, instead of a single surface like a field. This method often uses a controlled environment. medium.
- **Features:**
 - **Growing Medium:** It uses soilless agriculture, such as aquaponics, aeroponics, and hydroponics and receive nutrients through water or mist. Cultivators also commonly use coconut husk and peat moss as growing medium.
 - **Light source:** Cultivators use a combination of artificial and natural light, along with technologies like rotating beds to ensure optimum lighting efficiency.
- **Advantages:**
 - **Increased crop yield:** Controlled environment ensures optimum output in limited area.

- **Protection from weather disruption** as crops are generally placed indoor.
 - **Sustainability**: It minimizes use of natural resources. For e.g. it recycles irrigation water.
 - **Doesn't hamper local flora and fauna** as the land use is minimal.
 - **Urban farming**:
 - **Year around production** - as the crop is being grown in a controlled environment.
- **Limitations:**
- Large startup cost
 - Larger energy demands - due to dependence on supplementary lights like LEDs.
 - **Climate change** as currently fossil fuels emitting CO2 are the primary source of energy.
 - **Specialized technical expertise** is required for cultivating using this method.
- **Conclusion:** Vertical farming offers an alternative method of sustainable agriculture. To make it successful, its important to lower the initial cost, provide the required energy through renewable methods and provide skilling and training options to farmers willing to explore the option.

1) IMPORTANT CASE STUDY OF VERTICAL FARMING

- **Growing tomatoes without soil using Vertical Farming in Hydroponics**
 - » Vertical farming with hydroponics allows the cultivation of tomatoes without soil. It offers several advantages including space efficiency, reduced water usage, and year-round production.
 - » **Where has this method been implemented?**
 - Port Augusta Farm, South Australia. It features a 4.5 hectare greenhouse powered by a 51,500 m² concentrated solar power plant. This plant includes 23,000 mirrors that direct sunlight to a 127-meter-high tower weighing 234 tons.
 - » The generated heat serves three purpose:
 - **Maintaining optimal temperature** in the greenhouse
 - **Generating electricity** via a turbine to power farm systems
 - **Desalinating seawater drawn** from the nearby Spencer Gulf.
 - » **Outcome:**
 - The farm produces one million litre of fresh water daily by desalinating seawater from 3 kms away. It grows 7,000 tonnes of tomatoes annually (around 15% of Australia's total crop) on arid land. In addition, 1,80,000 tomatoes are grown hydroponically in stacks without soil, saving 2 million litres of diesel and reducing CO2 emission by 15,000 tonnes.
- **Understanding the method more:**
 - » The method entails growing plants in nutrient rich water solutions customized to their specific requirements. Vertical hydroponic systems can be employed in urban areas, reducing the environmental impact of transportation and offering locally sourced fresh produce.